

SLAVE LAKE, AB, Canada

WMO: 713690

Lat: 55.2964N

Lon: 114.7844W

Elev: 1912

StdP: 13.71

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-19.3	-33.5	0.9	-24.5	-27.7	1.3	-18.7	32.4	36.5	29.4	36.5	5.9	120	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	79.4	63.4	75.7	61.7	72.7	60.4	65.9	75.6	64.0	73.1	62.1	70.6	10.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	89.6	70.8	60.1	83.3	68.9	58.4	78.2	66.8	31.8	76.1	30.2	73.2	28.8	70.8	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.1	22.4	19.2	DB	-32.1	85.4	7.3	3.4	-37.3	87.9	-41.6	89.9	-45.7	91.8	-51.0	94.3	
			WB	-32.0	69.0	6.9	2.0	-36.9	70.4	-41.0	71.6	-44.8	72.7	-49.8	74.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	8.7	13.4	22.5	36.7	48.7	57.1	61.7	59.3	50.4	38.4	22.6	10.4	(6)
(7)		DBStd	21.51	15.98	12.88	13.67	8.74	7.64	5.15	4.63	5.35	6.84	7.58	11.52	14.13	(7)
(8)		HDD50	6178	1280	1026	854	404	115	6	0	3	78	364	821	1227	(8)
(9)		HDD65	10644	1745	1446	1319	850	506	242	123	187	440	823	1271	1692	(9)
(10)		CDD50	1046	0	0	0	4	75	218	362	292	89	6	0	0	(10)
(11)		CDD65	38	0	0	0	0	1	5	20	12	0	0	0	0	(11)
(12)		CDH74	495	0	0	0	3	35	77	208	144	28	0	0	0	(12)
(13)		CDH80	79	0	0	0	0	5	9	34	27	4	0	0	0	(13)
(14)	Wind	WSAvg	8.0	8.5	7.8	8.2	7.8	7.6	7.5	7.6	7.0	7.8	8.6	8.4	8.7	(14)
(15)	Precipitation	PrecAvg	20.7	1.1	0.7	0.8	0.8	1.7	3.6	3.9	2.7	2.0	1.1	0.8	1.1	(15)
(16)		PrecMax	25.9	2.7	1.7	2.3	1.7	4.3	7.3	7.8	5.4	5.2	2.8	1.8	2.4	(16)
(17)		PrecMin	13.7	0.3	0.2	0.1	0.0	0.6	0.4	1.6	0.2	0.7	0.1	0.0	0.2	(17)
(18)		PrecStd	3.2	0.6	0.4	0.5	0.6	0.9	1.9	1.6	1.4	1.1	0.7	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	44.8	51.0	70.1	78.5	80.6	83.8	83.5	77.9	65.5	50.8	46.4	(19)
(20)		MCWB	38.7	36.6	40.4	50.5	56.8	62.2	66.5	66.0	60.8	53.1	41.7	38.2	(20)	
(21)		2%	DB	41.2	40.0	45.7	59.9	72.2	75.8	79.6	77.9	71.6	57.7	44.3	40.6	(21)
(22)		MCWB	35.6	34.0	37.5	45.7	53.7	59.5	64.8	64.4	58.8	47.5	37.3	34.5	(22)	
(23)		5%	DB	36.6	35.8	42.6	54.1	67.5	72.1	75.6	73.8	66.9	53.9	40.6	35.4	(23)
(24)		MCWB	32.3	31.2	35.5	42.3	52.0	58.3	63.3	62.6	56.7	45.2	35.3	30.7	(24)	
(25)		10%	DB	30.7	31.8	39.7	49.6	63.3	68.9	72.4	70.5	62.8	50.5	37.0	30.2	(25)
(26)		MCWB	27.8	28.3	33.8	39.9	50.2	57.2	61.8	61.0	54.3	43.2	32.9	27.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	37.5	41.3	51.0	58.5	64.6	68.8	68.4	63.4	54.0	42.3	38.9	(27)
(28)		MCDWB	45.3	43.5	49.1	68.4	72.8	75.5	79.1	79.0	74.7	64.4	49.1	46.0	(28)	
(29)		2%	WB	35.8	34.5	38.4	46.0	56.0	62.0	66.4	65.9	60.1	48.8	38.7	34.8	(29)
(30)		MCDWB	40.7	39.3	44.9	57.8	68.5	72.0	76.2	75.4	69.5	55.7	43.5	40.4	(30)	
(31)		5%	WB	32.5	31.5	36.1	43.0	53.5	60.0	64.6	63.7	57.5	46.1	35.6	31.1	(31)
(32)		MCDWB	36.3	35.4	42.0	52.5	64.9	68.9	73.6	72.4	65.6	52.4	39.6	35.1	(32)	
(33)		10%	WB	27.8	28.2	34.0	40.5	51.2	58.1	62.7	61.8	55.2	43.7	33.2	27.5	(33)
(34)		MCDWB	30.7	31.9	39.1	48.5	60.8	66.4	71.0	69.6	61.8	49.8	36.6	30.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	16.0	18.5	18.5	18.6	21.0	18.5	17.9	18.5	18.7	15.4	13.3	15.1	(35)	
(36)		5% DB	MCDBR	21.3	23.3	20.9	28.2	29.0	26.4	24.9	25.1	27.5	23.2	17.5	22.3	(36)
(37)		MCWBR	16.6	18.3	14.0	16.6	14.7	13.7	13.0	13.9	16.3	14.8	12.7	17.3	(37)	
(38)		5% WB	MCDBR	20.8	22.6	19.8	25.6	25.5	21.7	22.4	22.9	25.2	21.0	16.1	21.7	(38)
(39)		MCWBR	16.4	18.0	13.5	15.6	13.5	12.3	12.6	13.2	15.5	14.1	12.1	17.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.247	0.261	0.289	0.333	0.353	0.362	0.367	0.364	0.319	0.288	0.254	0.230	(40)	
(41)		taud	2.377	2.442	2.456	2.406	2.401	2.387	2.423	2.385	2.532	2.584	2.468	2.332	(41)	
(42)		Ebn at Noon	229	269	285	283	283	280	276	269	271	257	226	208	(42)	
(43)		Edh at Noon	15	21	26	32	34	35	33	33	25	18	14	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	239	545	1009	1448	1767	1834	1827	1519	1042	567	261	168	(44)	
(45)		RadStd	22	50	98	120	153	136	109	115	98	74	26	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (23 neighbors)	N/A	N/A	+4.41	N/A	N/A	N/A	N/A	N/A	+99	N/A	(47)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A	

Nomenclature: See separate page